Aleksei Pshenichnyi

AI/ML Research Engineer · Generative Video · Technical R&D Lead

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Summary

Hands-on AI/ML engineer and technology entrepreneur with 15+ years of software engineering experience, specializing in generative video, audio-driven facial animation, and applied machine learning. Co-founder of Elai, an AI video generation startup acquired by Panopto in 2024. Experienced in end-to-end ML research and engineering, from neural model development and evaluation to production deployment and real-time inference optimization. Combines deep technical expertise with R&D leadership and a strong background in distributed systems and production software architecture. Authorized to work in the EU.

Skills

AI / ML: PyTorch, Deep Learning, Transformers, Convolutional Autoencoders, Generative Models, Computer Vision, LLM and Multimodal Model Applications

Specializations: Audio-Driven Facial Animation, Lip Synchronization, Neural Video Generation, Model Training, Evaluation Frameworks

Performance & Systems: Python, CUDA, Cython, Numba, JIT Optimization, Streaming Inference, Distributed Systems

Data Engineering: Kafka, ClickHouse, Cassandra, Greenplum, PostgreSQL, Scala, Java, Akka

Leadership: R&D Team Leadership, Research Planning, Technical Architecture, Mentoring

Experience

Panopto, Head of AI Research — Nov 2024 – present

Joined through Panopto's acquisition of Elai (October 2024). Lead R&D of the avatar video generation technology and new generative AI features, spending about 70% of my time hands-on on models, experiments and code.

- Developed and released a new generation of audio-driven facial animation models to production; platform users generated 300K+ avatar videos in 2024-2026.
- Researched expressive facial motion, addressing the over-smoothed articulation typical of supervised audio-to-animation models through training data design and architecture changes; integrated the results into the production models.
- Built and shipped to production an AI feature that generates animated visual content (charts) from text prompts or presentation slides, using LLMs to produce structured chart descriptions and multimodal models to interpret slides.
- Lead an R&D team of 4-5 engineers, combining research planning and technical architecture with direct implementation.

Elai, Co-Founder, Chief R&D Officer (Hands-on Principal Engineer) — May 2021 – Dec 2024

Co-founded Elai, an AI video generation startup acquired by Panopto in October 2024.

- Architected and contributed hands-on to production neural systems for audio-driven facial animation and realistic talking-head video generation, using transformer-based models and convolutional autoencoders in PyTorch.
- Led a 4-5-person R&D team while personally developing ML components, running experiments and making technical architecture decisions.
- Established model training, experiment tracking and evaluation workflows combining automated metrics with structured human assessments and preference testing.
- Developed a streaming real-time avatar MVP (LLM, text-to-speech, animation, neural rendering) achieving 25+ FPS and about 2-3 seconds from the first audio chunk to the first generated frame; the performance optimizations were reused in the main product.
- Optimized performance-critical components with CUDA, Cython, Numba and JIT compilation.

Eidolon, Head of Analytics (part-time) — Jan 2021 – Feb 2022

Designed and led development of an event analytics platform for mobile games.

- Designed the data models, processing workflows and reporting tools that turn user activity events into insights on engagement, conversion and product performance.
- Worked with game developers to apply the analytics in product decisions.

DataSpike, Technical Lead (Co-Founder) Oct 2019 – Jan 2021

Co-founded an early-stage startup building NLP-driven solutions for AML and KYC processes in financial institutions; led its research and development (Scala, Python, NLP, SPARQL).

Quadcode (IQ Option), Software Development (Data Engineering) Team Lead May 2017 – Oct 2019

Architect and hands-on lead of the analytics and ML data platform behind the IQ Option fintech product.

- Designed and built a distributed event analytics platform with reliable event delivery, deduplication and consistency verification; this platform and the later one at Eidolon sustained thousands of events per second (peaks of tens of thousands) on tens of TB of hot data.
- Cut recurring analytical reports from tens of minutes to seconds with pre-aggregations, partitioning and query plan optimization.
- Led ML systems built on the platform, including prediction of which new users become paying customers to optimize ad spend, fraud detection and trading news recommendations.
- Built real-time analytics and ML infrastructure with Scala and Akka and data pipelines with Python, Scala and Postgres.

Ice Owl, Software Developer (Backend & Game AI) Oct 2010 – Apr 2017

Developed server-side game logic, multiplayer features and AI for social and strategy games in Scala and Java.

- Designed memory-efficient planning for autonomous agents in a multiplayer tactical strategy game, reducing precomputed data from an estimated tens of GB to about 100 MB per map and enabling around 1,000 AI agents across hundreds of concurrent sessions on a single server.

Education

Computer Science Center, Data Analysis Saint Petersburg, Russia

- Evening postgraduate program (non-degree) Sept 2014 – June 2017

Siberian Federal University, Computer Science Krasnoyarsk, Russia

- Specialist degree (5-year, MSc equivalent) Sept 2004 – June 2009

Languages

English (Upper-intermediate (B2)) · Russian (Native) · Spanish (Basic (A1))